

RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8602 Compiler Design

Type of Activity: Think-Pair-Share

Topic: Intermediate Code Generation

Date: 17.02.20

Type of Learning:

Active Learning

Learning Objectives:

O3: To enable the students to understand intermediate code generation and syntax directed translation.

Activity Description:

Think-Pair-Share is a collaborative, active learning strategy.

- Initially the students were asked to think about the given topic individually without discussing with other in the class (Think).
- Each student is paired with another student.
- They share their thinking with each other and proceed with the task.
- Students share their thinking (or solution) with the entire class.

Uses of Think-Pair-Share:

- Students are actively engaged.
- Makes class interactive.
- It helps the students to think individually about a topic or answer to a given question.
- It helps to focus attention and engage students in learning

Justification for chosen the topic:

The Intermediate code Generation is one of the important topic, and repeatedly asked in university question. This activity makes the students to get a sound knowledge in this concept. Students think individually about a topic and share their ideas with their classmates that improve creative thinking and oral communication skills.

Implementation of Think-Pair-Share:

Think: The teacher posed questions to learn and explain the Intermediate code Generation. The students were grouped and discussed about the same. Some of the Questions are given below:

- | | |
|---|---|
| i) $a = b * c + b * d$ | ii) $\text{if}(a < b + c)$
$a = a - c$
$c = b * c$ |
| iii) $\text{while}(A < C \text{ and } B < D) \text{ do}$
$\text{if } A = 1 \text{ then } C = C + 1$
else
$\text{while } A \leq D$
$\text{do } A = A + B$ | iv) $\text{void main}()$
{
$\text{int } a;$
$a = 23$
$\text{if}(a == 23)$
$a = 10$
else
$a = 19$
} |
| v) $\text{if } A < B$
$\text{then } 1$
$\text{else } 0$ | |

Pair: The students were asked to discuss their pairs to understand the given concept as shown in Figure 1. The students discussed and learn by themselves or with their pairs to complete the task.

Share: Finally, the Five pair of students Aparna and Krishnaveni, Varsha and Muthulakshmi, Rathika and Ragapriya, Sudarshan Ramji and Prabhu, Muthurani and Gayathri, presented their concepts and participated in the discussion as shown in Figures 2-6.



Figure 1

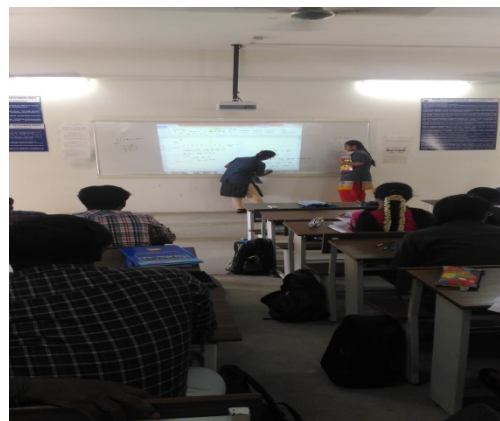


Figure 2



Figure 3



Figure 4

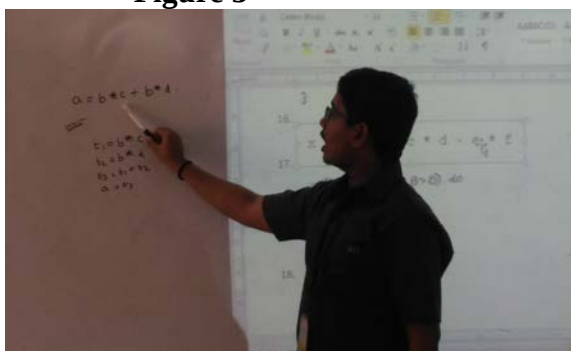


Figure 5



Figure 6

Outcomes:

- This activity increased attentiveness in the class and also improves the students' skill of critical analysis of the topics.
- This activity made the students to understand the topic clearly.

Relevance to POs

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
03	3	2	2					1	2	2		2

Reflective Critique:

Challenges

- Some students are hesitated and lack to share the concepts that they learnt in the class.
- Students found the activity helpful to get clarity on the topic especially when they combine and discuss.

Initiatives to address the problem:

- Several university questions were shown and discussed to the students about the weightage for intermediate code generation.
- Make the students to know the impact and importance of sharing their views and understanding related to the topic and made them involve in the activity.

Post-implementation

- Students were actively participated in this activity.
- From this activity, the students can get more clarity in the Intermediate code Generation by discussing and sharing their views with the other students in the class.
- The various representations of the intermediate code were again explained for their better understanding.

References:

https://www.ritrjpm.ac.in/images/computer-science/14_CS8392_TPS.pdf
https://www.ritrjpm.ac.in/images/computer-science/31_EC8393_TPS.pdf